Final Presentation — A DSA-Compliant Tourism RS

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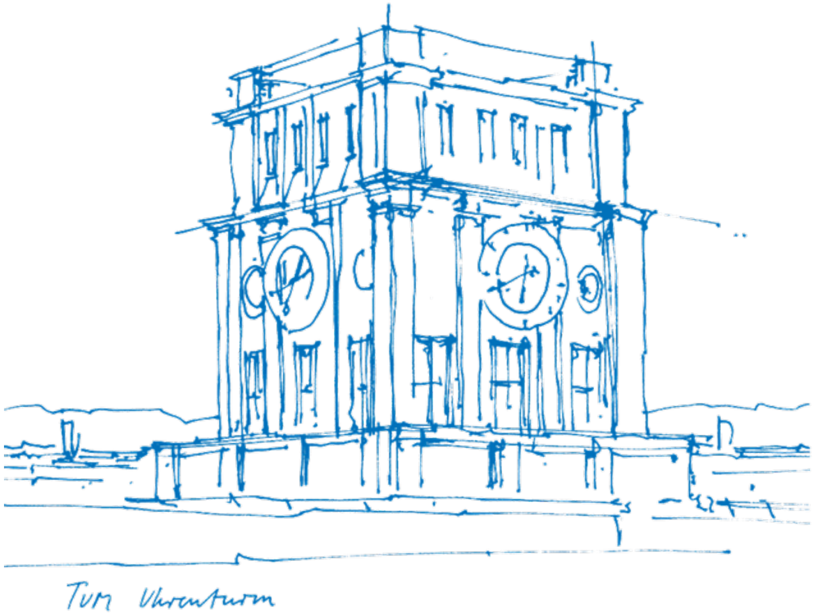

Background: Recommender Systems in Tourism

- **Unique Domain Characteristics [1, 2]:**
  - High financial risk and emotional investment.
  - Infrequent purchasing decisions (trips are planned only once or twice a year).
  - Multi-attribute search spaces (price, location, stars, transport, amenities).
- **Decision Complexity [1, 2]:**
  - Unlike low-cost retail products (e.g. movies, books), users do not buy travel packages on impulse.
  - Users require active comparison tools and want to express precise trade-offs.
  - Trust and perceived control over the algorithm are essential for user acceptance [3].

Motivation

- **The Recommender "Black Box:**
  - Most modern recommender systems (RS) hide their logic (e.g. neural network rankings) [4, 2].
  - Causes user distrust, especially in high-stake domains like tourism (travel planning, hotel bookings) [1, 3].
- **Regulatory Requirements: EU Digital Services Act (DSA) [5]:**
  - **Article 27:** Mandates that platforms explain the main parameters of their RS in plain, understandable language.
  - **Article 38:** Mandates that Very Large Online Platforms (VLOPs) offer at least one profiling-free recommendation alternative.
- **Project Goal:** Build an interactive, transparent & DSA-compliant tourism recommender system prototype.

Solution Multiple Algorithms including explanations

System Architecture & Technical Stack

- **Technology Stack:**
  - React[1] for component-based reactive state rendering.
  - Vite[2] as a fast bundler and build system.
  - Client-side KaTeX[3] rendering for mathematical formulas.
- **Real-Time Computation Flow:**
  - Eliminates the latency of database queries. Real-time updates occur client-side as users drag the sliders.
  - Caches the hotel calculations, avoiding unnecessary re-renders.
  - Normalization boundaries recalculate dynamically upon city changes.
- **Quality Assurance & Verification:**
  - Standalone recommender engine (`src/recommenderEngine.js`).
  - 60 automated unit tests (`run-tests.js`) verifying edge cases, boundary conditions, and preventing division-by-zero errors.

[1] `https://react.dev/`

[2] `https://vite.dev/`

[3] `https://katex.org/`

Two Algorithms with User Defined Weights/Limits

- **5 Attributes for Hotels:**
  - price per night, distance to center, stars, public transport access, user rating
- **Two Algorithms:**
  - **Weighted MAUT:** compensatory approach with user defined weights.
  - **Constraint-Guided:** Focused on hard limit rules and non-compensatory filtering.

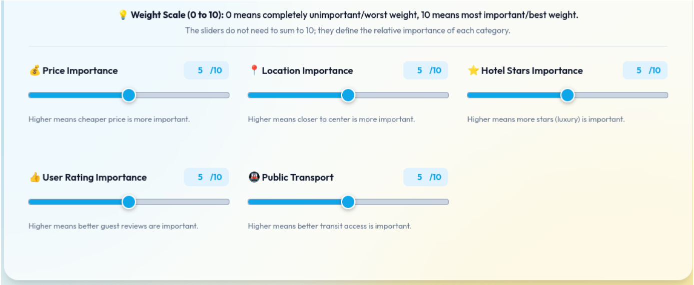

Abbildung: Dashboard

Demonstration Real-Time Strategy Toggling & Weight Adjustments

Algorithm A: Weighted MAUT (Compensatory)

- **Compensatory Utility Scoring [6]:**
  - High score in one attribute (e.g. location) can compensate for low performance in another (e.g. price).
  - Profiling-free (DSA-compliant)
- **Weights on a 0-10 Scale:**
  - Range: 0 (completely unimportant) to 10 (most important), changed after feedback in Milestone 3.
  - Mathematically normalized by the total sum of user-specified weights:

$$U = \frac{\sum w_i \cdot u_i}{\sum w_i} \times 100\%$$

(w_i: attribute weight [0–10], u_i: normalized utility [0–1], U: match score %)

Weighted Aggregation:

$$U = \frac{\sum_{i=1}^{5} w_i \cdot u_i}{\sum_{i=1}^{5} w_i} \times 100\%$$

Relative Normalization (Price & Distance):

$$u(v) = \frac{v_{\max} - v}{v_{\max} - v_{\min}}$$

Absolute Normalization (Stars & Review/Transport):

$$u_{\text{stars}}(v) = \frac{v-1}{4} \quad \text{and} \quad u_{\text{rating/transport}}(v) = \frac{v-1}{9}$$

Abbildung: Transparency Board Info

Normalization Strategies in Weighted MAUT

To aggregate attributes with different units, they are normalized to a $0-1$ scale [7]:

- **Relative Normalization (Cost attributes: Price, Distance):**
  - Normalizes value v relatively based on the active city's min/max values:

$$u(v) = \frac{v_{\max} - v}{v_{\max} - v_{\min}}$$

- Normalizing cost relatively maximizes the contrast between the local hotel options.
- **Absolute Normalization (Benefit attributes: Stars, Rating, Transit):**
  - Normalizes value v on fixed scale bounds as they are known:

$$u_{\text{stars}}(v) = \frac{v-1}{4} \quad \text{and} \quad u_{\text{rating/transport}}(v) = \frac{v-1}{9}$$

Algorithm B: Constraint-Based Filtering

- **Non-Compensatory Logic [1]:**
  - User defines absolute thresholds (e.g. maximum price, minimum stars) $\Rightarrow$ Hotel must satisfy all constraints.
  - Over-performing in one area cannot compensate for a single constraint violation.
- **DSA-Compliance:**
  - Transparent: Easy-to-understand inclusion/exclusion criteria.
  - Complete profiling-free predictability.

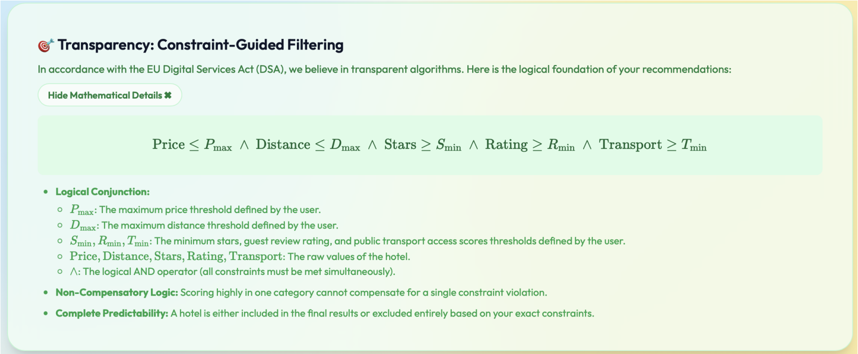

Abbildung: Constraint Filtering Interface

The Satisficing Model of Constraint Filtering

- **Logical Conjunction Formulations:**
  - A hotel is recommended if and only if:

$$\text{Price} \leq P_{\text{max}} \ \wedge \ \text{Distance} \leq D_{\text{max}} \ \wedge \ \text{Stars} \geq S_{\text{min}} \ \wedge \ \text{Rating} \geq R_{\text{min}} \ \wedge \ \text{Transport} \geq T_{\text{min}}$$

- **Psychological Background (Satisficing):**
  - Based on Herbert Simon's bounded rationality theory (1957).
  - Decision-makers do not seek the mathematically optimal hotel, but rather a "good enough"hotel that meets all minimal conditions.
  - Drastically reduces cognitive load during selection.
  - Perfect for users with rigid conditions (e.g., maximum budget).

Improvement upon Milestone Feedback

Initial Usability Issues:

- Implicit strategy selection (no clear entry point).
- Entering Point showed overloaded transparency board before seeing the sliders.
- Sliders scaled from $0 - 100\%$, confusing users into thinking weights must sum to 100%.

Applied Redesign:

- **Interactive Strategy:** landing page.
- Importance scale changed to **0-10 range** to prevent percentage-summing confusion.

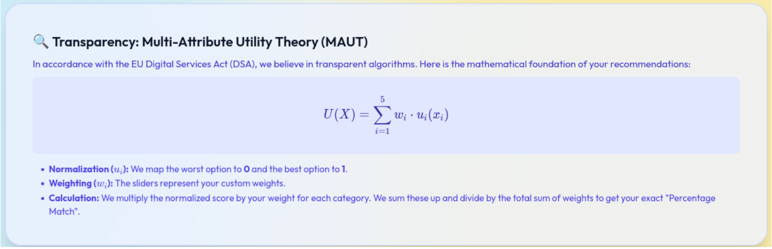

Abbildung: Previous Cluttered Transparency Board

Different Explanation Levels

- **The Explanation Dilemma [8]:**
  - Giving too little explanation makes the system look like a black box.
  - Giving too much explanation causes cognitive overload and user fatigue.
- **Tiered Disclosure Features:**
  - **Novices:** Brief text explanations focusing on the meaning of user controls. Satisfies general transparency.
  - **Experts (KaTeX Formulas):** Button shows raw equations, especially good for transparency audits and regulators.
  - **Interested Users:** Collapsible Step-by-step breakdown $\Rightarrow$ Explains individual recommendations, boosting user trust.

Future Work: Proposed Hybrid Recommender

Addresses individual limitations of MAUT and Filtering:

- **Limitation of Weighted MAUT:** Compiles complete lists; does not filter out completely irrelevant options (e.g. extremely expensive hotels).
- **Limitation of Constraint Filtering:** Screens items but does not rank remaining candidates (higher cognitive effort).

Proposed Two-Stage Hybrid Approach:

1. **Stage 1: Constraint Filtering (Non-Compensatory screening):** Filter out hotels violating basic requirements (e.g. price $>$ \$250).
2. **Stage 2: Weighted MAUT (Compensatory ranking):** Apply preference weights to rank remaining candidates.

$\Rightarrow$ Maintains complete mathematical transparency, remains profiling-free, and mimics realistic human decision-making.

Proposed User Study & Evaluation Design

- **Hypotheses:**
  - **H1 (Perceived Control)**
  - **H2 (System Trust)**
  - **H3 (DSA Compliance Impact)**
- **Within-Subjects Laboratory Study ($N = 30$):**
  - **Condition 1 (Baseline):** Traditional black-box hotel list.
  - **Condition 2 (Weighted MAUT):** Slider controls with collapsible breakdowns.
  - **Condition 3 (Constraint Filtering):** Threshold sliders.

Summary & Conclusion

- **Successfully Completed Deliverables:**
  - Developed an interactive, responsive tourism recommender SPA (React + Vite).
  - Redesigned selection flow with a strategy landing page to prevent early transparency board clutter.
  - Configured sliders on an intuitive 0-10 scale to eliminate weight-summing misunderstandings.
  - Integrated tiered explainability panels (plain-language summaries, LaTeX formulas, math breakdowns).
- **Key Takeaway:**
  - Demonstrates that EU DSA compliance (Article 27 & 38) can be harmonized with rich user controls, mathematical transparency, and premium UI/UX aesthetics.

Thank you for your attention!

Questions?

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